AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 4 · STUDENT LESSON

Activation functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For z = 2 times width - height - 3 at [3, 3], what are the score and class under z >= 0?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace identity ReLU and sigmoid on a numerical score. Try the recall before opening notes.

Your learning target

I can trace identity **ReLU** and **sigmoid** on a numerical score.

I can explain how a nonlinear activation changes a computation.

I can distinguish an activation **output** from a final class or verified probability.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace identity **ReLU** and **sigmoid** on a numerical score

Explain how a nonlinear activation changes a computation

Distinguish an activation **output** from a final class or verified probability

Retrieve what you already know

> For z = 2 times width - height - 3 at [3, 3], what are the score and class under z >= 0?

My starting explanation

This week's end goal

By the end, I can explain **activation function**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Activation functions case. Separate observations, given facts and assumptions before choosing an explanation.

First compute an affine score z = w dot x + b. An activation then returns a = f(z). In a network this value may become an **input** to another layer; it is not automatically a class **prediction**.

We will compare three functions using the same incoming scores so that only the activation changes.

Identity returns z unchanged. **ReLU**, pronounced ree-loo, returns the larger of zero and z. Thus **ReLU**(-2) = 0, **ReLU**(0) = 0 and **ReLU**(2) = 2. Positive scores keep their magnitude; **ReLU** does not turn every positive value into 1.

Incoming z	Identity output	ReLU output
-2	-2	0
0	0	0
2	2	2

These are fixed arithmetic functions. Selecting a function or changing its **input** is not itself a **training** procedure. Later weeks will introduce losses and parameter updates.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Activation functions. Connect each labelled input, deciding step and result.

Sigmoid is 1 / (1 + exp(-z)). Here exp(t) means e raised to power t, where e is approximately 2.718. A calculator or the supplied Python code can evaluate it; no calculus is required this week.

For z = -2, 0 and 2, **sigmoid output**s are approximately 0.1192, 0.5 and 0.8808.

For every finite mathematical **input**, **sigmoid** is strictly between 0 and 1, though computer rounding can produce an endpoint for extreme scores.

Negative **input**s produce values below 0.5, zero gives 0.5, and positive **input**s give values above 0.5. A class rule still needs to be stated: if class 1 means **sigmoid**(z) >= 0.5, these three **prediction**s are 0, 1 and 1.

```
from math import exp

def sigmoid(z):
    if z >= 0:
        return 1 / (1 + exp(-z))
    e = exp(z)
    return e / (1 + e)

for z in [-2, 0, 2]:
    print(z, max(0, z), round(sigmoid(z), 4))
```

The two algebraically equivalent branches avoid taking exp of a large positive number. A value within [0, 1] is not automatically a reliable probability: its interpretation depends on the fitted task and evaluation **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Activation functions. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Replace **ReLU** with identity.

Predicted result

Prediction: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Why the result follows

Reason: Identity retains the negative intermediate value instead of clipping it.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Activation functions to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Activation function	A function applied to a computed score
ReLU	The function max(0, z), retaining positive scores and replacing negative scores with zero
Sigmoid	The function 1 / (1 + exp(-z)), mapping finite real scores into the interval between 0 and 1
Nonlinear function	A function that cannot be represented by one fixed slope and offset over its entire domain

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Activation functions, predict a step, then compare with its explanation.

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

For z = -1 and z = 1, give identity and ReLU outputs. Using sigmoid(-1) = 0.2689 and sigmoid(1) = 0.7311, give classes under the rule a >= 0.5.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Activation functions

See what **ReLU** removes before the next weighted step.

The facts we will use

Supplied facts

Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/4

ReLU hidden computation at x=1

Input 1 x weight 1 = 1

Add the products and bias -3: total = -2

ReLU: use max(0, total) -> output 0

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/4

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute preactivation	z values are -2, 0, 1.
2. Apply ReLU	h values are 0, 0, 1; the negative value is replaced by0.
3. Compute output	y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

What this example does not establish

Keep the stated limit

ReLU is one activation choice. Clipping negatives does not itself train the network or prove

useful representation.

Keep the stated limit

Read the labelled picture: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Replace **ReLU** with identity. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Activation functions claim and explain a repair using the supplied evidence.

Check the claim against the facts

An affine function has one fixed slope. **ReLU** has slope zero on the negative side and slope one on the positive side, so no single straight line fits the whole function.

For a concrete check, **input**s -2, 0, 2 give **output**s 0, 0, 2. The average of the endpoint **output**s is 1, but the **output** at the midpoint 0 is 0. This differs from an affine function, which preserves midpoints.

Check the claim against the facts

For a one-**input** chain with identity between stages, h = 2x + 1 followed by y = 3h - 2 collapses to y = 6x + 1. Insert **ReLU** instead: h = max(0, 2x + 1), y = 3h - 2.

At x = -1, the identity chain gives -5 while the **ReLU** chain gives -2. At x = 1, both give 7. The nonlinear step can change the overall family of functions.

Locate the earliest unsupported step

Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Activation functions with support, then explain the deciding step in your own words.

For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.

For z = -1 and z = 1, give identity and ReLU outputs. Using sigmoid(-1) = 0.2689 and sigmoid(1) = 0.7311, give classes under the rule a >= 0.5.

Explain your prediction to a partner before comparing results.

Practise with less help: Apply a declared class rule

Use workbook W2 for the connected example "**Activation function**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Activation function**s. Proposed explanation: "**ReLU** returns a class label". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Activation functions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

```
from math import exp

def sigmoid(z):
    if z >= 0:
        return 1 / (1 + exp(-z))
    e = exp(z)
    return e / (1 + e)

for z in [-2, 0, 2]:
    print(z, max(0, z), round(sigmoid(z), 4))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Activation functions. Explain why each step is justified.

Explain → apply → challenge

For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.

Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.

A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **activation function**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace identity **relu** and **sigmoid** on a numerical score.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A nonlinear activation allows a network to represent relationships that a chain of affine operations alone cannot express. It does not guarantee that **training** will find a useful model or that more layers will help.

ReLU can erase differences between negative scores; **sigmoid** can compress very large scores into nearly identical values. The architecture, data and **training** procedure still matter.

> **A learner reports sigmoid(2) is about 0.8808 and concludes the system will be correct 88 percent of the time for every future case. What does the calculation establish, and what evidence is still missing?**

Next week, we will compose layers while keeping every intermediate value and its role visible.

A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

> **My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Activation functions case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Trace a new chain independently

Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Activation functions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: See what **ReLU** removes before the next weighted step.

By the end, I can	How I will show it
Trace identity ReLU and sigmoid on a numerical score	Explain the deciding step, show a trace or test, and state one limit.
Explain how a nonlinear activation changes a computation	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an activation output from a final class or verified probability	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Activation functions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace identity **ReLU** and **sigmoid** on a numerical score

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See what **ReLU** removes before the next weighted step.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.

1. Apply **ReLU**

2. Compute **output**

3. Compute preactivation

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compute **output** y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. **ReLU** means keep a positive value and replace a negative one with zero.

The **output** layer receives that transformed value, not the original **input**. **Input**s one and three therefore both deliver zero to the final layer and produce two.

Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace **ReLU** with identity. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. h becomes -2, 0, 1 and **output**s become -4, 2, 5.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: h becomes -2, 0, 1 and **output**s become -4, 2, 5. Identity retains the negative intermediate value instead of clipping it.

Repair a missing building block

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

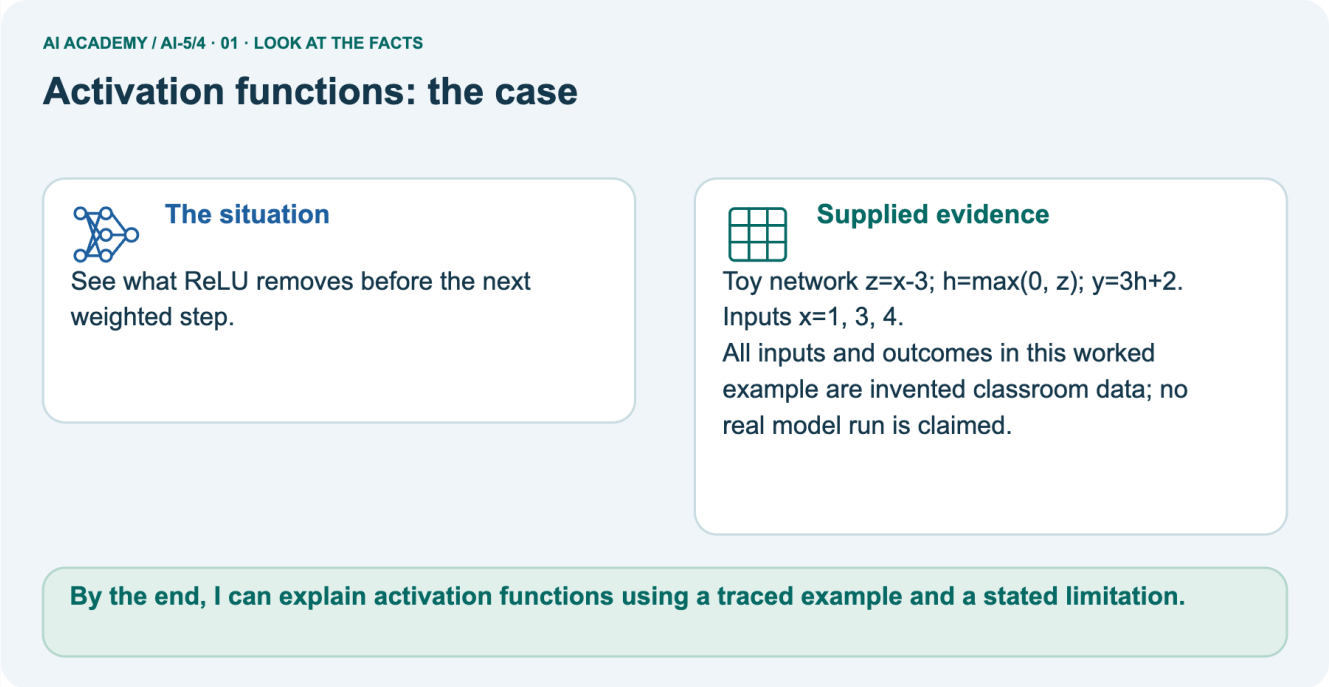
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Activation functions graphic using its labels, deciding evidence and relevant limitation.

01. Activation functions: the case



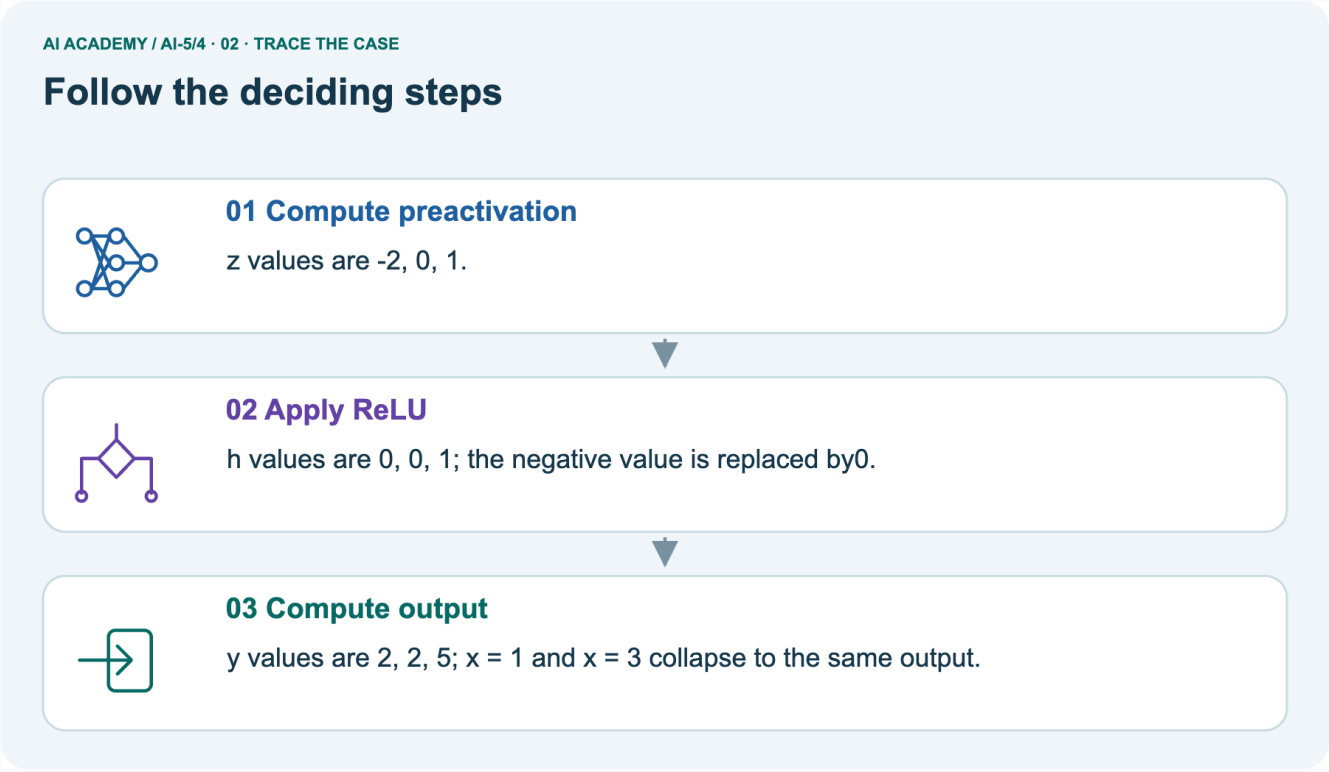
Follow the labelled picture with the matching explanation. Open the vector diagram

See what **ReLU** removes before the next weighted step. Supplied facts: Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

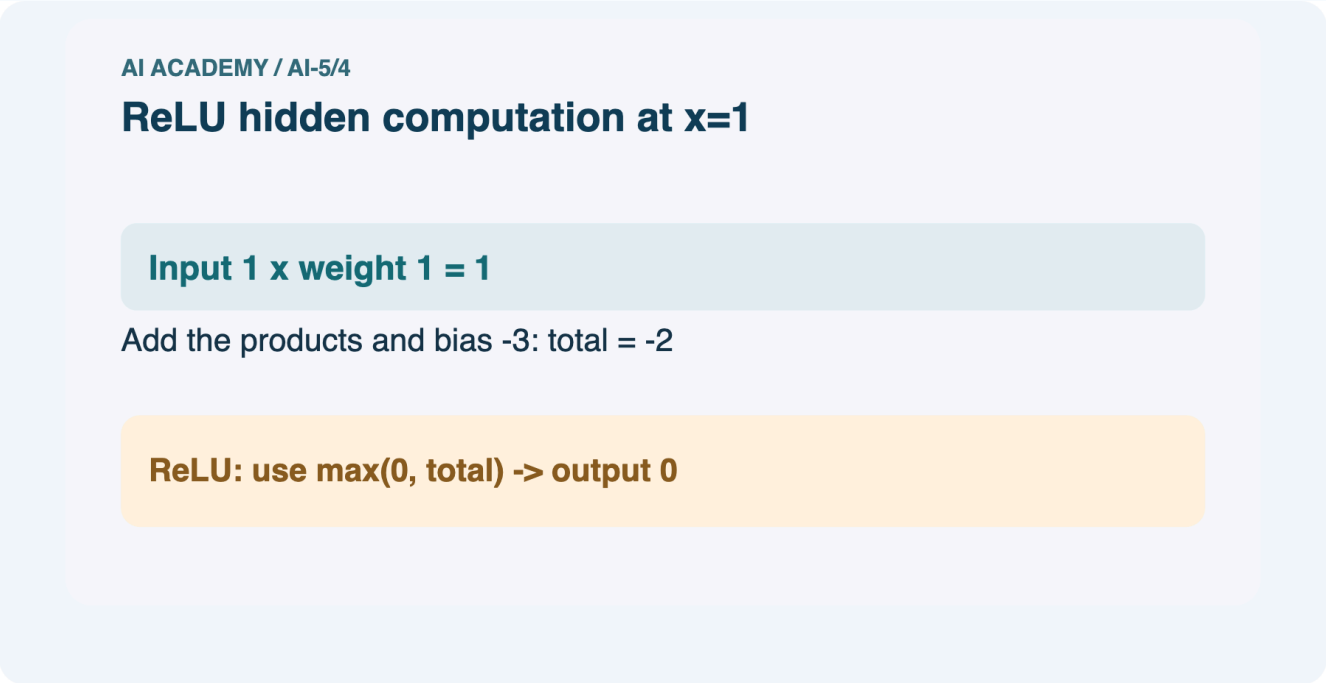


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute preactivation: z values are -2, 0, 1. → Apply **ReLU**: h values are 0, 0, 1; the negative value is replaced by0. → Compute **output**: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.

> Point to each numbered stage and explain which recorded **input** it uses.

03. ReLU hidden computation at x=1

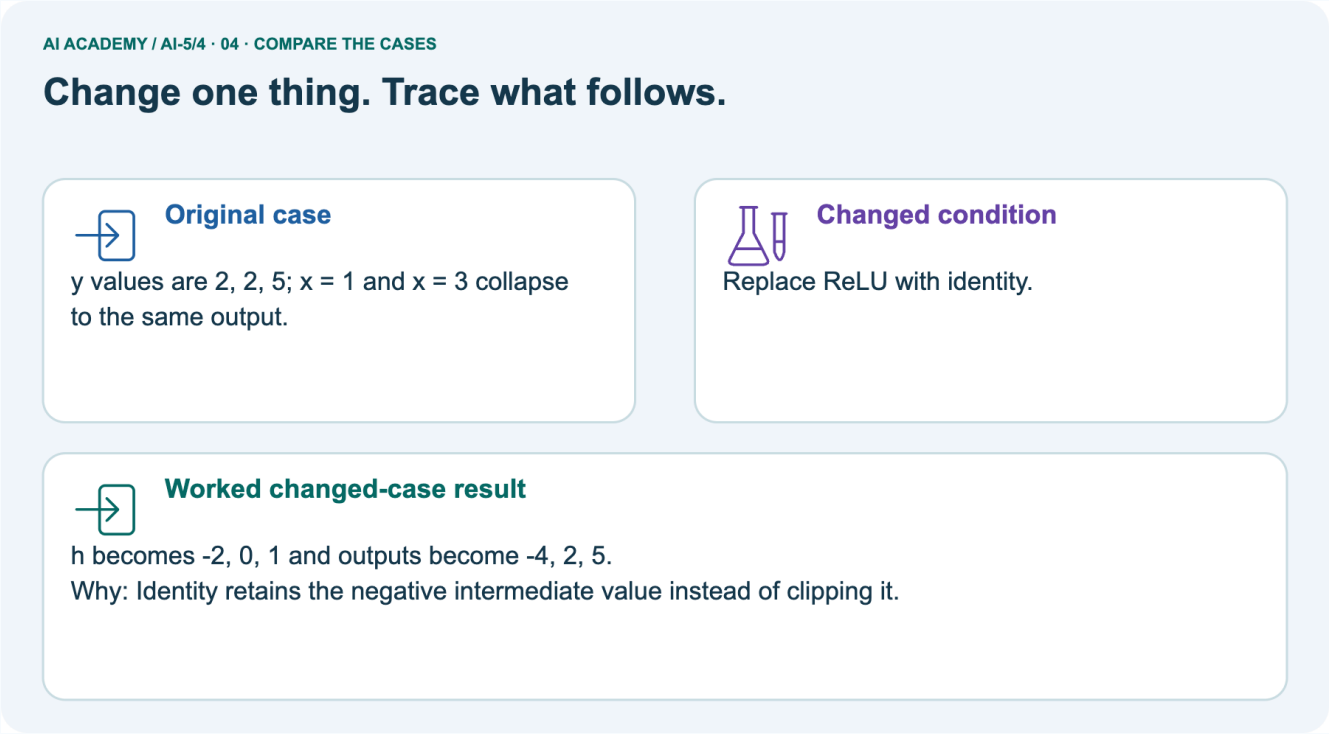


Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU hidden computation at x=1. **Input**s: [1]; weights: [1]; bias: -3; activation: **relu**; **output**: 0.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



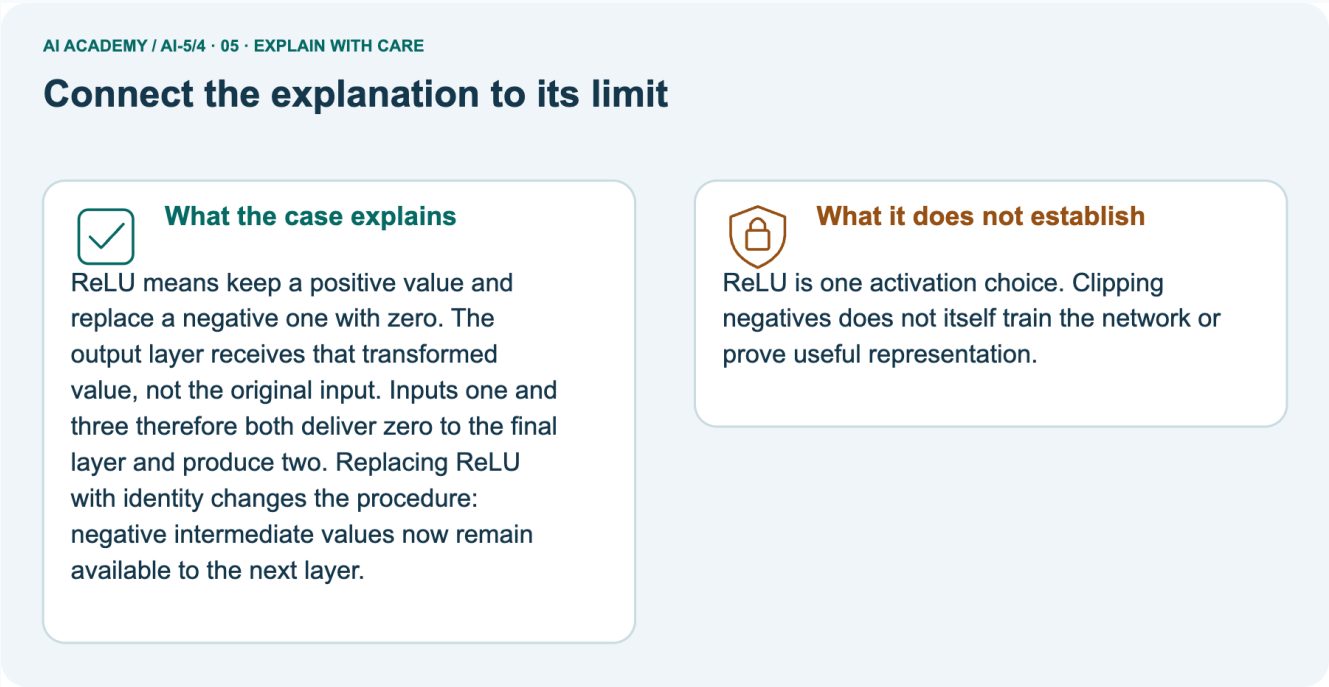
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. Changed condition: Replace **ReLU** with identity. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

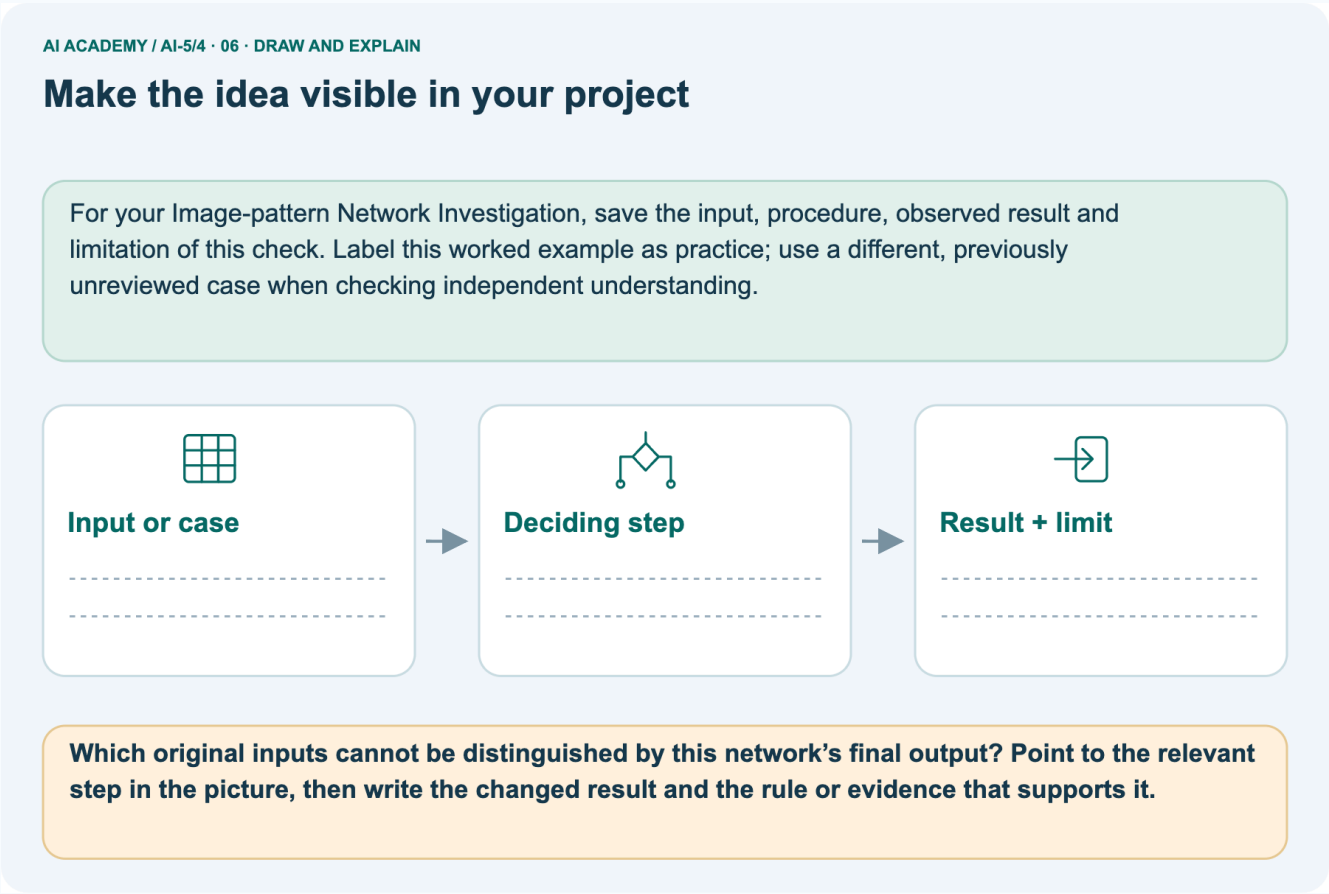
ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Limit: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.